

Shubhan Mehrotra

Software Engineer | Backend Engineer

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Education

Degree/Certificate	Institute/Board	CGPA / Percentage	Year
B.Tech. (Electrical Engineering)	Indian Institute of Technology Palakkad	7.85	2025
Senior Secondary	CBSE Board	90.8%	2021
Secondary	CBSE Board	93.8%	2019

Experience

Entrupy

Software Engineer I

Jun 2025 – Ongoing
Bengaluru, India

- Recognized with Q3 Growth Mindset Award and Q4 Think Big Award for driving continuous learning, strategic problem-solving, and business-aligned technical execution.
- Eliminated correlated per-row subqueries in a cold-cache analytical query by rewriting it into a parallelized two-pass execution pipeline (fast bucket-count pass plus a separate correction pass, combined at the application layer), slashing p99 endpoint latency by over 95% for large organizations.
- Architecting a custom, event-driven payment recovery engine using AWS Lambda, SQS FIFO queues, and Event-Bridge to replace rigid third-party retry systems with dynamic, error-classified smart retries.
- Architected and deployed a server-side Role-Based Access Control (RBAC) framework, replacing legacy implicit authentication with secure JWT token-exchange to enforce granular, row-level data isolation.
- Developing scalable FastAPI backend services for a customer-facing analytics dashboard serving 10K+ daily users, building high-performance endpoints across multiple microservices to support real-time business operations.
- Architected an automated Post-Invoice Reconciliation Engine using Python and AWS S3 to perform session-level diffs between analytics and snapshots, ensuring 100% financial accuracy.
- Engineered a multi-tenant billing framework for 200+ accounts, automating contractual minimums and tiered-pricing recovery to eliminate revenue leakage and reduce overhead by 30%.
- Built event-driven monitoring and subscription tracking via AWS Lambda, SQS DLQs, and Chargebee webhooks, ensuring 100% message reliability for 50+ monthly plan upgrades.
- Optimized system performance and reliability by designing DynamoDB GSIs to reduce API latency and refactoring exception handling to raise test coverage from 80% to 88%.
- Integrated CRM and Slack APIs with backend services to automate ticket routing for 15+ exception types and manage 90-day credit expiry, saving 10+ hours of manual triage weekly.

Entrupy

Python Developer Intern

Oct 2024 – May 2025
Bengaluru, India

- Designed and deployed feature flags and access control systems across 4 backend microservices, replacing static role checks and reducing manual access updates by 50%.
- Engineered an automated dual-profile assignment system with an admin UI, reducing user onboarding time by 50%.

TCS Research

Computing Systems Research Intern

Jun 2024 – Jul 2024
Thane, India

- Curated and preprocessed 50GB+ of datasets for RAG-based training on LLaMA and Mistral architectures, improving data throughput by 35%.
- Optimized hardware utilization metrics and model data flows, reducing distributed training time by 27.5% on high-compute clusters.

Projects

Phoneix – Self-Healing Distributed Cache & Personal API

Personal Project

May 2026 – Present
[Live Demo](#) | [GitHub](#)

- Engineered a distributed cache in Go from scratch featuring consistent hashing (150 virtual nodes/peer), gossip-based failure detection (~2s window), and automatic replication with zero-downtime rebalancing.

- Built a FastAPI backend and vanilla JS dashboard on ARM64 Linux, serving real-time host metrics, GitHub activity, and system logs via custom `/proc` filesystem parsers.
- Architected an edge reverse-proxy setup using Nginx, Cloudflare Tunnels, and Supervisor to route traffic across multi-process Go/Python microservices running on a mobile Linux node.

Pixel 7a Bare-Metal Linux Edge Server

May 2026 – Present

Personal Project

[GitHub](#)

- Repurposed an ARM64 Android device into a 24/7 headless Debian Linux edge server running GrapheneOS and Magisk root, achieving a low-power (~3–5W idle) self-hosted infrastructure.
- Configured zero-trust public routing via Cloudflare Tunnels and Dropbear SSH over WiFi, bypassing CGNAT and eliminating static IP requirements.
- Automated containerized service deployments using Docker Compose (Nginx, Vaultwarden, Uptime Kuma) while managing strict memory limits on a 6GB RAM node.

Agentic RAG & Retrieval Evaluation Platform

Dec 2024

Inter IIT Tech Meet 13.0

- Built an agentic RAG system for technical documentation with dynamic query routing across simple, intermediate, and complex retrieval agents using LLM-based query classification.
- Developed adaptive quality controls using GuardRails and relevance thresholds to determine retrieval quality and when to augment responses with web search.
- Evaluated retrieval quality using RAGAS across context precision, recall, faithfulness, and answer relevancy, and benchmarked re-ranking strategies using NDCG@10 and MRR@10.
- Built resilient retrieval infrastructure with pluggable vector-store indexing and fallback web-search APIs for improved reliability.

Ollive – LLM Inference Observability Platform

July 2026

Personal Project

[GitHub](#)

- Built a full-stack LLM observability platform (Django/DRF backend, React frontend) that auto-instruments inference calls via decorators requiring zero per-call-site logging code, streaming responses to the client over Server-Sent Events.
- Designed an event-driven ingestion pipeline (HTTP → Kafka → consumer → ClickHouse) decoupling log storage from the request path, with PostgreSQL for OLTP conversation state and ClickHouse for OLAP-style analytics on inference logs.
- Built a real-time analytics dashboard surfacing throughput, average/p95 latency, and error rate, with regex-based automatic PII redaction (emails, phone numbers, SSNs, card numbers, IPs) applied before log transmission.
- Deployed via Docker Compose for local development and Kubernetes manifests for production.

Technical Skills

Languages: Python 3, Go, TypeScript, SQL (PostgreSQL, MySQL), Java, C/C++

Backend: Django, FastAPI, RESTful API Development, Microservices Architecture, Billing Systems

Frontend: React Native

Databases: MySQL, PostgreSQL, DynamoDB

Cloud & DevOps: AWS (ECS, SQS, Lambda, IAM, CloudWatch, EC2), Docker, CI/CD Pipelines

Data & Analytics: SQL Optimization, ETL Pipelines, CSV/Pivot Reports **Tools:** Git, GitHub Copilot, Claude Code, Cursor, Graphviz, PostHog, HubSpot, Chargebee, Slack API